

CHE 2201 – Physical Chemistry II

Phase Equilibria

Lec. 4. Phase Diagrams and Distillation of Binary Systems

4.1. Introduction

In general, the distillation of industrial oils, such as petroleum oils, oils from natural products are carried out at constant pressure (i.e. atmospheric pressure). Constant pressure distillation is cheaper and convenient than constant temperature distillation.

Using Raoult's law and Dalton's law of partial pressures, the vapour pressure of a two component system for any composition in the liquid phase at const. T , can be calculated (Refer to section 3.2). This information can be used to draw graphs of vapour pressure versus mole fraction in the liquid phase or in the vapour phase. Usually, in these systems, the vapour phase is rich in more volatile components and the liquid phase is rich in low volatile components.

4.2. Phase Diagram of a Binary System

In the diagrams we have discussed during the previous lecture (See figures 3.1 to 3.4), the total vapour pressure of a binary system was shown as a function of the composition of the solution (i.e. the liquid phase). Since the composition of the vapour phase differs from that of the liquid phase, we can also use the composition of the vapour phase as a variable in the diagram. In this way, we can obtain a second curve of the same property of a system (i.e. total vapour p), depending on a different variable (composition of the vapour phase). This leads to the phase diagram of a binary system (liquid-vapour equilibria).

4.2.1. Constant Temperature Condition

The figure 4.1 represents a plot of vapour pressure versus mole fraction of a two component system A and B at constant T .

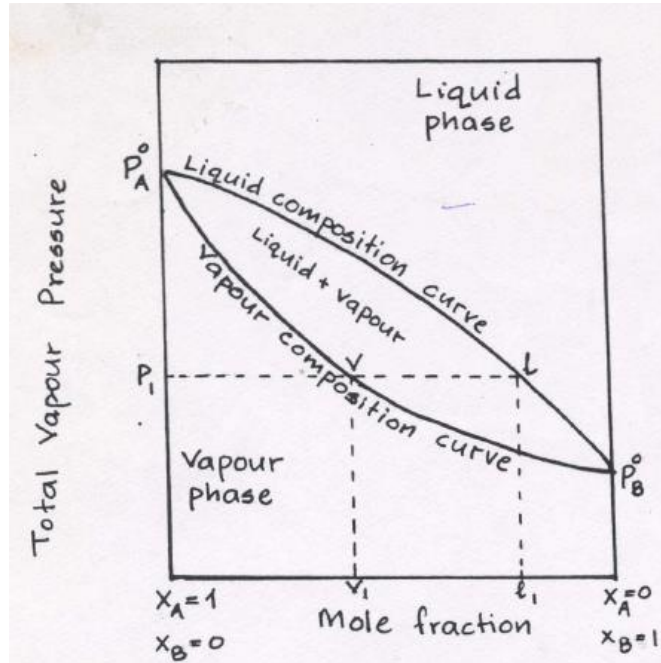


Fig. 4.1. Liquid-Vapour Equilibria at Const. Temperature

Here, we have considered component A to be more volatile than B, i.e. $p_A^0 > p_B^0$. As such, at a given p , the vapour must consist more of A than B. The upper curve (liquid composition curve) shows how the vapour p depends on the composition of the liquid, and the lower curve (vapour composition curve) shows how it depends on the composition of the vapour.

These curves divide the plane of the diagram into three parts. The upper part corresponds to the existence of a single liquid phase (i.e. solution of a variable composition). The lower part corresponds to a gas mixture of variable composition. Thus, any point in the upper and lower parts shows the state of one existing phase. The area enclosed by the two curves represents the coexistence of vapour and liquid in equilibrium (two phases).

Since there are two phases enclosed in this area, namely liquid and vapour, we can calculate the degrees of freedom using the phase rule.

$$F = C + 2 - P \quad (4.1)$$

$$P = 2; C = 2;$$

$$F = 2 + 2 - 2 = 2.$$

Thus, the system processes two degrees of freedom, and it is bivariant. In other words, at const. T , the system is defined by p and X of the mixture. This means, at a given p , the composition of the liquid and the vapour are fixed.

For the areas above liquid composition curve and below vapour composition curve, only one phase is possible, i.e. liquid and vapour phase respectively.

$$P = 1; C = 2;$$

$$F = 2 + 2 - 1 = 3.$$

It means that the system has three degrees of freedom; p , T and X . So, at const. T , the system depends only on p and X . Such a system is isothermally bivariant.

The compositions of vapour and liquid phases at any point can be determined by using the curves in Fig. 4.1. According to this figure, at a vapour pressure of p_1 , the composition of liquid is l_1 , and the composition of vapour is v_1 ($v_1 \neq l_1$). The line vl is called the tie line, because it connects (or ties together) the two phases in equilibrium.

4.2.2. Constant Pressure Condition

Distilling liquids under conditions of const. T is not a widely used technique. Most separations are carried out at const. external pressure, usually at normal atmospheric pressure. In this case, we have to determine the normal boiling points for various compositions of the mixture, and then draw a graph of T vs. X . Similar to the const. T condition, two curves can be drawn in this case too. One of these is a boiling temperature versus the mole fractions of components in the liquid phase, and the other is the boiling temperature versus the mole fractions in the vapour phase.

This diagram is different from that of p vs. X , because here the vapour phase exists above the curves (i.e. at high T s), and the liquid phase is the one that is below the curves (i.e. at low T s). The enclosed area represents the coexistence of vapour and liquid phases. In this case, the normal boiling temperature of A (T_A) is less than that of B (T_B), and as such, the vapour phase is rich in A than B.

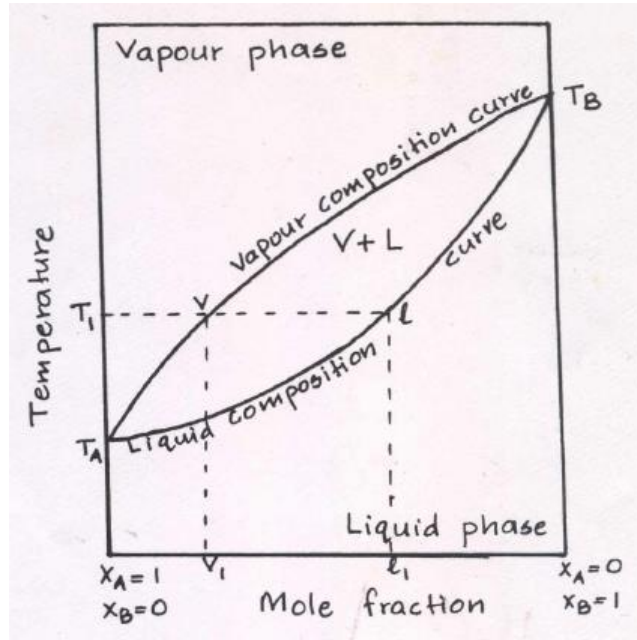


Fig. 4.2. Liquid-Vapour Equilibria at Const. Pressure

Similar to the previous phase diagram (Fig. 4.1), this diagram can also be used to determine the compositions of liquid and vapour phases at any given point. For example, at temperature T_1 , the composition of liquid is l_1 , and the composition of vapour is v_1 . The line vl is called the tie line.

4.3. Distillation

Distillation is actually the collection and condensation of vapour of a binary liquid, where the collection takes place in a different part of the apparatus from the boiling process. As mentioned earlier, this can be carried out either at const. T or at const. p . In industries, most of the distillation processes are carried out at const. p (isobaric) conditions, since it is more convenient and cheaper than the const. T (isothermal) condition.

4.3.1. Isothermal (Const. T) Distillation

At constant T , two miscible liquids exist in equilibrium with its vapour. If the vapour in the system is removed, more of the liquid will vapourize. Since the more volatile liquid vapourizes easily, the remaining liquid in the residue will consist of more of less volatile liquid. The continuous distillation will result the pure component of the less volatile

liquid as the residue, and the more volatile component as the distillate. This procedure of separation at const. T is known as the isothermal distillation.

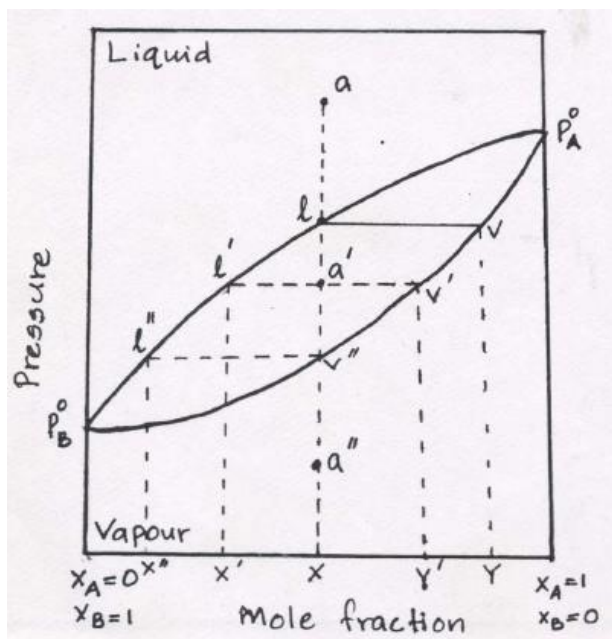


Fig. 4.3. Isothermal Change in Pressure with Mole Fraction

Let us consider the point 'a' in Fig. 4.3. At this point, the system is entirely liquid, and it remains so as the pressure is reduced until the point 'l' is reached. At 'l', the first trace of vapour appears having a composition Y (line lv). Since A is more volatile than B, the vapour of composition Y is richer in more volatile A.

As the pressure is reduced further, the system moves from point l to point a' . During this reduction, the composition of the liquid moves along the line ll' , and the composition of the vapour moves along the line vv' . At point a' , the liquid composition is X' , and the vapour composition is Y' .

Further reduction of pressure brings the system to the point v'' . At this point, only a trace of liquid of the composition X'' remains, and the vapour has the composition X. Reduction of pressure down to the point a'' results in the same composition as the original liquid X.

This discussion shows that the vapour formed over the liquid becomes richer in a particular component than the liquid. This is used as the basis for the separation of

components from liquid mixtures. However, the isothermal distillation can be used to separate only ideal or near-ideal solutions.

4.3.2. Isobaric (Const. p) Distillation

As mentioned earlier, isothermal distillation is not a convenient procedure, and used only if other methods are not appropriate. In general, distillation processes are carried out at const. p, as it is convenient and easy to control the conditions of the system.

The normal boiling point of a mixture is the temperature at which the total vapour pressure becomes one atmosphere. Therefore, the distillation is usually carried out at 1 atm. The boiling point of the solution varies with composition. The curve which represents this variation is called the boiling point curve.

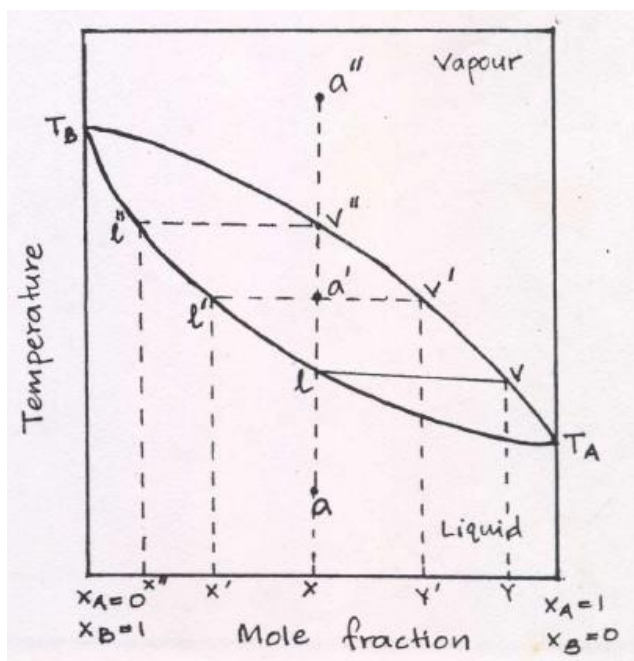


Fig. 4.4. Isobaric Change in Temperature with Mole Fraction

Let us study the sequence of events that occur during the process of heating a mixture at point 'a' to a higher temperature a'' . At point 'a', the system consists entirely of liquid. As the T rises, the system remains as liquid until the point 'l' is reached. At this temperature T_l , the first trace of vapour appears having the composition Y. Here, the vapour is much

richer in A, since the normal boiling point of A is less than B. This is the basic fact for the separation of volatile mixtures by distillation.

As the T is raised upto the point a' , the liquid composition changes along the line ll' , and the vapour composition changes line vv' . At a higher T, at point v'' , the last trace of liquid composition X'' disappears. And above the point v'' , the system exists entirely in the vapour phase.

4.3.3. Repetitive Distillation

The simple distillation process (i.e. one stage process), can be used to separate the components, which have large difference in their boiling Ts. However, in the majority of cases, the components will have similar boiling Ts, and as such, one stage distillation is ineffective. Separation of such liquids can be greatly improved by repeating one stage process several times. This can be understood by studying the phase diagram of benzene-toluene system.

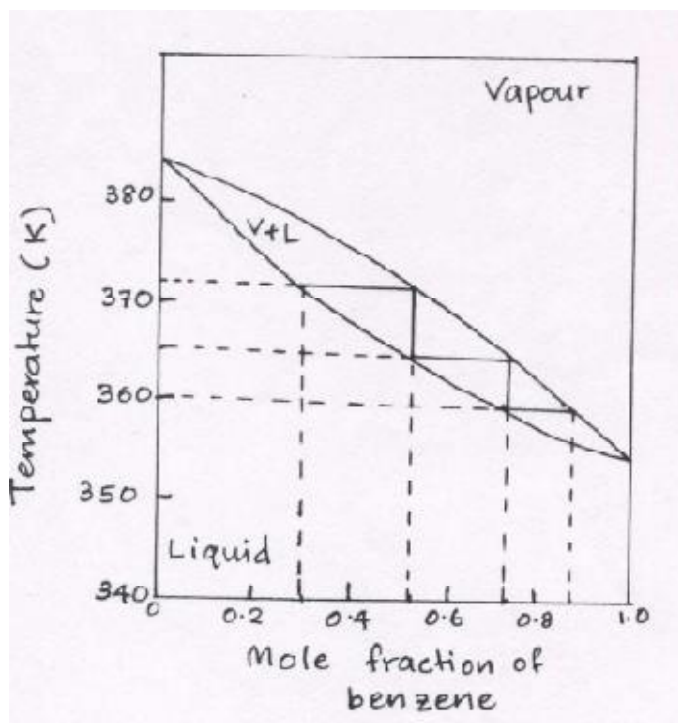


Fig. 4.5. Phase Diagram for Benzene/Toluene System at $p = 1 \text{ atm}$

The liquid mixture with 0.71 mole fraction of benzene boils at 365 K. The vapour formed at this T is collected and condensed, the resulting liquid is transferred to a separate boiling flask, and the process is repeated. Now the liquid will boil at 360 K, and the composition of the first vapour formed will contain 0.85 mole fraction of benzene (See Fig. 4.5). After a few such distillations, the vapour collected in the final receiver will have a mole fraction of benzene approaching 1.0, i.e. pure benzene.

But this method of separation (repetitive distillation) is ineffective on economic grounds, as it is a process of repeating the basic apparatus. The most efficient technique is known as the fractional distillation.

4.3.4. Fractional Distillation

Fractional distillation is performed in a column, known as the fractionating column. This is shown diagrammatically in the Fig. 4.6.

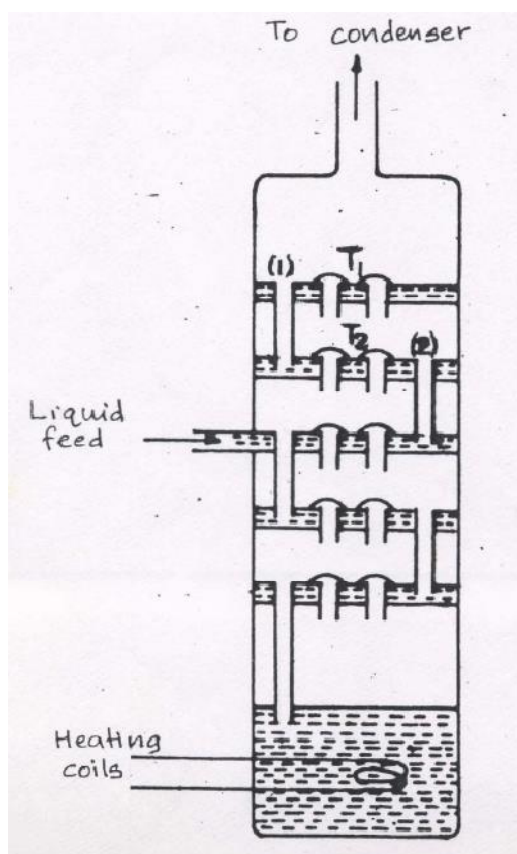


Fig. 4.6. Schematic Fractional Distilling Column

Each plate in the column is equivalent to a condenser and a boiling flask. It condenses the vapour from the plate below, and act as a flask for the enriched liquid. The number of plates required for separation can be determined from the phase diagram of the mixture to be separated.

For example, in order to separate a mixture containing 0.85 mole fraction of benzene, from a mixture containing 0.30 mole fraction of benzene (See Fig. 4.5), we will need an equivalent of three theoretical plates. Since the first distillation occurs between the flask and the base of the column, the actual number of plates required is two. This gives the minimum number of plates required in practice.

As we can see from the Fig. 4.6, the column is heated at the bottom. There is a temperature gradient along the length of the column, the top being cooler than the bottom.

If we consider the top plate 1, the vapour in that column is in equilibrium with the liquid in plate 1, and compositions of liquid and vapour are l_1 and v_1 respectively. Similarly, in plate 2, the compositions are l_2 and v_2 , and its temperature is higher than that of plate 1. As the vapour in the plate 2 passes upward to plate 1, it is then cooled to the temperature of plate 1 (T_1), and some of the vapour (v_2) condenses to form l_1 .

Since the liquid in top plate is richer in less volatile component, the remaining vapour is richer in more volatile component, and at equilibrium attains the composition v_1 .

This occurs at every plate in the column, i.e. the liquid phase is in equilibrium with vapour. As the vapour moves upwards in the column, it cools and condenses the less volatile component. Thus, the vapour consists more of higher volatile component as it moves upwards, and the lower plates are richer in less volatile component. If the individual components have a large difference in boiling points, a distilling column with few plates is sufficient to separate the mixture, and vice versa.